Muhammad Huzaifa

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Summary

MS student in Robotics & Autonomous Systems at KFUPM with a research background in applied machine learning, computer vision, and embedded systems. Author of a peer-reviewed publication on markerless, vision-based physical assessment, with additional research experience in signal processing and antenna design. Previously worked as an AI Engineer at Isentia, developing LLM-based conversational AI for media monitoring.

Education

King Fahd University of Petroleum & Minerals (KFUPM) — Dhahran, Saudi Arabia
M.S. (Project-Based) in Robotics & Autonomous Systems — *Aug. 2026 – Present*

National University of Sciences and Technology (NUST) — Islamabad, Pakistan
Bachelor of Engineering in Electrical Engineering — *Nov. 2021 – June 2025*

Army Public School and College (APS&C) — Rawalpindi, Pakistan
Higher Secondary Education, Pre-Engineering – 97.36% — *Apr. 2019 – July 2021*

Experience

AI Engineer — 2025 – Present
Isentia (placed with client PulsarGroup Ltd.) — *Remote*

- Develop and deploy LLM-based conversational AI chatbots for intelligent media monitoring, from pipeline design through integration with real-time content-analysis workflows
- Build end-to-end AI/ML pipelines connecting large language models to media-monitoring data sources to support automated, large-scale content analysis
- Operate as part of a fully remote engineering team embedded with the client (PulsarGroup Ltd.), delivering production features under its engineering and security standards

Research & Hands-On Experience

Research Intern – Intelligent Field Robotics Lab (NCAI) — June 2024 – June 2025
National University of Sciences and Technology (NUST) — *Islamabad, Pakistan*

- Built a non-contact frailty assessment system for the elderly using Kinect V2 and computer vision
- Applied machine learning on joint-tracking data for real-time frailty classification
- Improved model accuracy and deepened expertise in CV-based health analysis

Research Intern – Adaptive Signal Processing Lab (SINES) — June 2024 – Sept. 2024
National University of Sciences and Technology (NUST) — *Islamabad, Pakistan*

- Developed a motor fault detection system combining machine learning, control systems, and cloud integration
- Applied signal processing to sensor data for real-time fault analysis, improving detection accuracy and system reliability

Research Intern – TUIL Lab (RIMMS) — June 2023 – Dec. 2023
National University of Sciences and Technology (NUST) — *Islamabad, Pakistan*

- Studied and designed a 7-element circularly polarized antenna
- Designed and fabricated single-element, four-element, and seven-element antennae using HFSS Ansys
- Improved isolation between elements and achieved higher gain across antenna configurations

Research Intern – Optical Networking and Technologies Lab (SINES) — June 2023 – Sept. 2023
National University of Sciences and Technology (NUST) — *Islamabad, Pakistan*

- Implemented network communication protocols using Cisco Packet Tracer and open-source network technologies
- Researched and analyzed communication protocols among network nodes
- Strengthened skills in network design and protocol analysis

Machine Learning Intern – ByteWise Summer Internship — June 2024 – Sept. 2024
ByteWise — *Islamabad, Pakistan*

- Completed a 100-day internship implementing machine learning algorithms and models to solve complex problems

Final Year Research Project

Frailty Assessment for the Elderly | *Kinect V2, Computer Vision, Machine Learning* – IRL, NCAI — June 2024 – June 2025

- Led a fully funded project in collaboration with Juntendo University Graduate School of Medicine to develop a non-contact, computer-vision-based frailty assessment system
- Built a user-friendly application for real-time analysis and reporting, achieving up to 100% classification accuracy

Research Publications

A Markerless Vision-Based Physical Frailty Assessment System for the Older Adults — 2025

- Huzaifa, M.*; Ali, W.; Iqbal, K.F.; Ahmad, I.; Ayaz, Y.; Taimur, H.; Shirayama, Y.; Yuasa, M. *AI* 2025, 6, 224. IF: 5.0

Projects

Face and Eye Detection System | *Python, OpenCV, Tkinter* GitHub

- Built a facial detection engine using pre-trained Haar classifiers with OpenCV, with a Tkinter GUI for real-time face and eye detection

Plant Disease Identification | *Deep Learning, MobileNetV2, Grad-CAM* GitHub

- Classified and visualized plant leaf diseases from the PlantVillage dataset using MobileNetV2 and Grad-CAM, reaching ~90% accuracy

Fault Detection of Induction Motor | *Machine Learning* GitHub

- Built an ML-based system to detect induction motor faults (rotor, bearing, winding) by analyzing sensor data for anomalies

CKD Diagnosis | *Machine Learning, Logistic Regression, SVM, Random Forest* GitHub

- Built a Chronic Kidney Disease diagnosis model by classifying patients with Logistic Regression, SVM, and Random Forest

Brain Tumor Detection Using CNNs | *Deep Learning, CNNs* GitHub

- Classified brain tumors from MRI scans with a CNN, using image preprocessing, data augmentation, and model optimization to boost accuracy

Real-Time OCR with Deep Learning | *Python, EasyOCR* GitHub

- Built a deep-learning OCR system extracting text from a live camera feed, tested for robustness under extreme lighting and poor visibility

Technical Skills

AI/ML: PyTorch, TensorFlow, OpenCV, Pandas, NumPy
Languages: Python, C/C++, Java, MATLAB
Infrastructure: Docker, Kubernetes, Minikube
Hardware Description: Verilog, VHDL
Tools: MATLAB, Proteus, Altium Designer, Adobe Suite, Office Suite, Visual Studio
Microcontrollers: STM32, ESP32, ATmega, Arduino, Raspberry Pi

Certifications

AI in Healthcare Specialization (Certificate) • Introduction to Machine Learning (Certificate) • Applied AI with Deep Learning (Certificate) • Introduction to TensorFlow for AI, ML and Deep Learning (Certificate) • MATLAB Onramp (Certificate) • Introduction to Artificial Intelligence (IBM) (Certificate)

Languages

English: Fluent – IELTS 8 Bands
Urdu: Native / Fluent
German: Beginner

Volunteer Experience

Social Volunteer, Pehli Kiran School, Islamabad
Social Welfare Volunteer, Red Crescent Society of Pakistan

Related Coursework

Digital Signal Processing • Signals and Systems • Computer and Communication Networks • Engineering Network Analysis • Machine Learning • Electrical Machines • Power Systems Analysis • Power Systems Protection